Kaylee Y. Vo

kayleeyvo@gmail.com | 650.798.7237 | San Francisco, CA | LinkedIn | Website

EDUCATION

Harvard University
Master of Liberal Arts (ALM) in Extension Studies, Data Science | GPA: 4.0

Stanford University
Bachelor of Arts in Economics (BA) | *Minor: Statistics*

WORK EXPERIENCE

Harvard School of Engineering and Applied Sciences, *Cambridge, MA* *Sept '25 – Present*
AI Research Assistant
- Neural operators: Under the guidance of Professor Pavlos Protopapas, I conducted experiments on various model architectures, including DeepONet and Fourier Neural Operator, to develop a one-shot model for ODEs and PDE solutions.
- Physics-Informed Neural Networks: Working on a novel neural network that can solve PDEs and is agnostic in the domain. The current model applies a Vec2Vec translation, which translates the latent representations from one domain to another, allowing the model to generalize. We optimize using Newton's method instead of gradient descent for the final linear layers.

Nuna Inc, *San Francisco, CA* *Mar '23 – Jun '24*
Software Engineer - Data Engineering
- Productionized the Chronic Illness and Disability Payment System (CDPS) model. The model was deployed to EMR Spark clusters, using Prefect for task orchestration. Optimized the CDPS algorithm, resulting in an 80% reduction in runtime.
- Built data transformations to map external client data to Nuna's data schema. Schemas are defined using Avro files and written to an AWS Glue. Performed Spark optimizations to reduce the ETL runtime. Wrote unit and integration tests to ensure data quality. The work culminated in a production Glue database used across the company.

Pngme, *San Francisco, CA* *Nov '21 – Nov '22*
Data Scientist - Data Platform
- As the sole Data Scientist, spearheaded all Pngme's data initiatives. Created quarterly roadmaps and OKRs, planned bi-weekly sprints, and collaborated with internal and external stakeholders for GTM strategy and product feedback. Managed two reports (data science contractors) in addition to individual contributor work.
- Developed an MVP loan default prediction model, a logistic regression fitted on cross-sectional data. Features are financial metrics transformed into weight-of-evidence features using the OptBinning library. Lift, profit curves, and Shapley values were used as monitoring metrics, and AUC was the performance metric.
- Lead feature engineering and modeling R&D work to ensure ML model performance increases commensurately with data growth. We generated custom word embeddings with BERT using user SMS data as input. The final model was a mixture of experts, with a logistic regression meta-model. The final model was customer-facing and deployed to production.

Toast Inc., *Boston, MA* *Nov '18 – April '20*
Data Scientist - Toast Capital
- Built a new line of business (LOB) called Toast Capital. The product comprises three patented models: LSTM Revenue Prediction (#11562425), a probability of default model (#11100575), a dynamic pricing engine (#10956974), an ETL data pipeline, and API endpoints. Total origination >$15 million, the second most profitable line of business.
- Designed prototypes to improve the PD model. This includes model selection, model stability, model parsimony, TTC PD vs PIT PD, and testing regression assumptions. I also led an R&D initiative to develop a generalized PD model, which can generate PD estimates for any arbitrary time horizon.

SKILLS AND AWARDS

Computer Languages:	Python (Sklearn, PyTorch, Tensorflow, JAX), R, SQL (Postgres, Snowflake, BigQuery), bash, MATLAB
Mathematics and AI:	Vision (semantic segmentation, R-CNNs, ViT), LLM (LLaMA, BERT), Diffusion (U-Net, StableDiffusion), Multimodal (CLIP), Stochastic & Partial Differential Equations, Real & Functional Analysis, Matrix Theory, Causal Inference, Time Series.
Enterprise:	AWS (Sagemaker, S3, EC2, EMR, Glue), Prefect, Docker, Git/Github Actions, Kubernetes, Spark
Awards:	Stanford University Academic Scholarship (Full-Ride), Stanford Economics Student of the Month (10/15/15 – 12/15/15), Grant Recipient for Economics Research, Haas Center Research Grant Recipient, Center on Poverty & Inequality Certificate (1st Undergraduate Research Fellow), President of Stanford Economics Association